Seokweon Jung

Postdoctoral Researcher

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Research Interest

My name is Seokweon Jung, and I am currently a Postdoctoral Researcher at the Data Interaction Group, HCII at Carnegie Mellon University. I received my Ph.D. in Computer Science and Engineering from Seoul National University, advised by Prof. Jinwook Seo at the Human-Computer Interaction Lab. My research expertise spans Visual Analytics, Network Analysis, and Human-AI Collaboration. Currently, I focus on developing innovative methodologies that integrate visualization and machine learning to facilitate the analysis of complex datasets across diverse domains.

Education

Mar.2019–Aug.2025 **Ph.D. in Computer Science & Engineering**, *Seoul National University*, Seoul, Republic of Korea.
Ph.D. Thesis: When Dynamic Graphs Meet Human: Visual Analytics Approach

Mar.2012–Feb.2019 **B.S. in Computer Science & Engineering**, *Seoul National University*, Seoul, Republic of Korea.

Job Experience

Mar.2026–Feb.2027 **Visiting Scholar in Data Interaction Group, HCII**, *Carnegie Mellon University*, Pittsburgh, United States.

Sep.2025–Feb.2026 **Postdoctoral Researcher in LLM Innovation Research Center**, *KAIST*, Daejeon, Republic of Korea.

Publications

Journal Articles

2026 **Seokweon Jung**, Jeongmin Rhee, Seoyoung Doh, Hyeon Jeon, Ghulam Jilani Quadri, and Jinwook Seo. Seeing graphs like humans: benchmarking computational measures and mllms for similarity assessment. *arXiv preprint arXiv:2602.22416*, 2026. To be presented in VisMeetsAI Workshop colocated with PacificVis 2026, and to be published in Information Visualization.

2023 **Seokweon Jung**, DongHwa Shin, Hyeon Jeon, Kiroong Choe, and Jinwook Seo. Monetexplorer: A visual analytics system for analyzing dynamic networks with temporal network motifs. *IEEE Transactions on Visualization and Computer Graphics*. IEEE, 2023.

2023 **Seokweon Jung**, Donghwa Shin, Jinwook Bok, Seokhyeon Park, Hyeon Jeon, Jinwook Seo, Insoo Lee, and Sooyoung Park. Interactive visual analytics system for criminal intelligence analysts with multiple coordinated views. *Journal of KIISE*, volume 50, pages 47–56, 2023.

Conference Papers

2026 Suyeon Hwang, Minkyu Kweon, Jeongmin Rhee, Soohyun Lee, Seokhyeon Park, **Seokweon Jung**, Hyeon Jeon, and Jinwook Seo. Hooklens: Visual analytics for understanding react hooks structures. *arXiv preprint arXiv:2602.17891*, 2026. To be presented in PacificVis 2026.

2024 Minsuk Chang, Donghun Kim, Hyeon Jeon, **Seokweon Jung**, and Jinwook Seo. Clever: Continual learning visualizer for detecting task transition failure. In *2024 IEEE 17th Pacific Visualization Conference (PacificVis)*, pages 369–374. IEEE, 2024.

2022 Sebeom Park, Soohyun Lee, Youngtaek Kim, Hyeon Jeon, **Seokweon Jung** Jinwook Bok, and Jinwook Seo. Vant: A visual analytics system for refining parallel corpora in neural machine translation. In *2022 IEEE 15th Pacific Visualization Symposium (PacificVis)*, pages 181–185. IEEE, 2022.

2021 **Seokweon Jung** Kiroong Choe, Seokhyeon Park, Hyung-Kwon Ko, Youngtaek Kim, and Jinwook Seo. Mixed-initiative approach to extract data from pictures of medical invoice. In *2021 IEEE 14th Pacific Visualization Symposium (PacificVis)*, pages 111–115. IEEE, 2021.

2021 Kiroong Choe, **Seokweon Jung** Seokhyeon Park, Hwajung Hong, and Jinwook Seo. Papers101: Supporting the discovery process in the literature review workflow for novice researchers. In *2021 IEEE 14th Pacific Visualization Symposium (PacificVis)*, pages 176–180. IEEE, 2021.

Workshop Papers and Posters

2026 Jiyeon Bae, Mingyu An, Jeongin Park, **Seokweon Jung**, Kiroong Choe, and Jinwook Seo. Physical containers as framing conditions for visualization in augmented reality. In *19th IEEE Pacific Visualization Conference (PacificVis)*, 2026. (Poster).

2026 **Seokweon Jung**, Jiwon Song, Yumin Song, Jinwook Seo, and Ha-kyung Hidy Kong. A scoping review on how hci researchers visualize results of thematic analysis. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2026.

2025 **Seokweon Jung**, Hyeon Jeon, Jeongmin Rhee, and Jinwook Seo. Can vlms assess similarity between graph visualizations? *18th IEEE Pacific Visualization Conference (PacificVis)*, 2025. (Poster) Best Poster Award.

2024 Kiroong Choe, Seokhyeon Park, **Seokweon Jung**, Hyeok Kim, Ji Won Yang, Hwajung Hong, and Jinwook Seo. Supporting novice researchers to write literature review using language models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pages 1–9, 2024.

2024 **Seokweon Jung**, DongHwa Shin, Hyeon Jeon, and Jinwook Seo. Combinational nonuniform timeslicing of dynamic networks. In *17th IEEE Pacific Visualization Conference (PacificVis)*, 2024. (Poster).

2021 **Seokweon Jung** and Jinwook Seo. Comparison of node-link diagram and adjacency matrix visualizing network with a group of nodes which have no interconnections. In *Proceedings of HCI Korea 2021*, pages 800–803, 2021. (Poster).

2019 **Seokweon Jung** and Jinwook Seo. A visual analytics application for rebalancing bike-sharing systems. In *Proceedings of KCC 2019*, pages 1400–1402, 2019. (Poster).

Research Experience

Seoul National University, Republic of Korea

Aug.2024 ***Research Workshop***.
In SNU-CMU Collaboration Workshop Summer 2024, we share knowledge and research experiences about Cognitive, Interactive, and Societal AI with faculties and students of SNU and CMU.

Sep.2018 – Feb.2019 ***Undergraduate Intern***.
With the aim of developing a novel interaction without using hands, we built a machine that can recognize users' heads tilted and serve spices. The result was submitted and accepted to the 2019 ACM CHI Student Research Competition.

Advisor : **Prof. Jinwook Seo**, *Associate Professor, Department of Computer Science & Engineering*, Seoul National University (*Homepage*)

Yonsei University, Republic of Korea

Jun.2018 – Aug.2018 ***D.REAM ACADEMY***.
Participated in the D.REAM ACADEMY's 2018 Summer Program for Companion, hosted by Yonsei University's Human and Robot Interaction Center. Designed a chatbot service to help people interact with AI for dementia care.

Teaching Experience

Fall, 2024 **M1522.006600: Intelligent Computing System Design Project (Human-AI Interaction System)**, *Seoul National University*, Teaching Assistant.

Spring, 2021 **4190.426A: Human Computer Interaction**, *Seoul National University*, Teaching Assistant.

Funded Grants

Sep.2024 – Aug.2025 **Research Grant for Ph.D. Students**, *National Research Foundation of Korea*, RS-2024-00410700.
Visualization of Anomaly Detection in Cryptocurrency Transactions with AI and Nonuniform Network Slicing.(Amount: 25,000,000₩ $\simeq$ 18,000$)

Talks

Nov. 2025 **Javascript-based Opensource Software for Visual Analytics of Network Data**, *Kwangwoon University, Seoul, Republic of Korea*.

Academic Services

Conference Reviewer (External)

2025–2026 IEEE Visualization and Visual Analytics (VIS)

2025–2026 ACM Conference on Human Factors in Computing Systems (CHI)

2024–2026 IEEE Pacific Visualization Conference (PacificVis)

Student Volunteer

2024 IEEE Visualization and Visual Analytics (VIS)

2023 IEEE Pacific Visualization Symposium (PacificVis)

Fellowships & Awards

Apr. 2025 **Best Poster** | *18th IEEE Pacific Visualization Conference (PacificVis 2025)*

Mar. 2025 **Star Student Researcher** | *Brain Korea 21 (BK21) Four Program*

Apr. 2022 **Best Visualization Notes** | *15th IEEE Pacific Visualization Symposium (PacificVis 2022)*

Computer skills

Programming Languages: Python, PyTorch, C, C++, JAVA

Web Technologies: Javascript, Typescript, React, Svelte

Database: SQL, MySQL

References

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HCI Lab
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Dr. Adam Perer
Associate Professor, Human-Computer Interaction Institute
Data Interaction Group
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